

INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR

BE407: Introduction to Cell Biology BE407

Course structure

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Lectures: Tuesday and Friday 3.30 pm - 4.50 pm (7/207)

Instructor: Dr. Subramanian Sankaranarayanan

Office: AB6-407D

Pre-requisites: ES243 Biology for Engineers or BE 101 Introduction to Life Sciences: Fundamentals of Life

Course contents:

Module a. **Origin of Cell Biology** (Week 1 & 2)

History and development of Cell Biology; Invention of microscopes; Discovery of cells; Cell theory; Membrane and bulk phase theories; Steady-state membrane pump concept

Module b. **Cell structure and organization** (Week 3 – 5; **Quiz 1**)

Cell as a unit of life, prokaryotic and eukaryotic cells, difference between plant and animal cells, bio-membranes, Structure and functions of various cell organelles i.e., nucleus, ribosome, ER, golgi, lysosomes, peroxisome, exosomes, vacuoles, chloroplasts, cytoskeleton; Organization of tissues: Cell-cell and cell matrix interactions, cell adhesion molecules, composition of extracellular matrix, cellular junctions and role; Cellular movement: types of movement, cilia and flagella, muscle contraction, role of cytoskeletal proteins in movement

Module c. **How cells obtain energy from food** (Week 6 -7; **Assignment 1**)

Glycolysis, Citric acid cycle, Energy generation in mitochondria and chloroplasts, oxidative phosphorylation, electron transport, photosynthesis

Week 8 – Exam 1

Module d: **Cell cycle, Control of Cell Numbers and Cell Size** (Week 9-10)

G1, G2, S and M phase of the cell cycle, Cell cycle analysis and its applications, Programed cell death apoptosis vs necrosis, Mitogens stimulating cell division, growth factors

Module e: **Transport systems in cells** (Week 11-12; **Quiz 2**)

Osmosis, active and passive transport, protein transporters, ion channels, protein sorting, vesicular transport, secretory pathways, endocytic pathways, exocytosis, micropinocytosis, role of cytoskeletal proteins in transport, molecular motors

Week 13 – Mid semester Recess

Module f: **Cell signaling** (Week 14-15, **Assignment 2**)

Principles of cell signaling, Concept of receptor-ligand, Signaling molecules, Receptor types: ion-channel-coupled receptor, G-protein Receptor, Enzyme coupled

receptors

Module g: **Techniques used in cell biology and relevance of cell biology** (Week 16-17)

Histology, staining, microscopy, TEM and SEM, live cell imaging, Fluorescent dyes and genetically encoded fluorescent tags, cell culture, FACS; Understanding Cancer, Stem cells and application, Tissue engineering, Infectious diseases, manipulating cells for therapeutic purposes

Week 18 – Exam II

Reference Books:

1. Alberts. Bray. Hopkin. Johnson. Lewis. Raff. Roberts and Walter. Essential Cell Biology. Fourth Edition. Garland Science. Taylor and Francis group
2. Alberts. Johnson. Lewis. Morgan. Raff. Roberts. Walter. Molecular Biology of The CELL. Sixth Edition. Garland Science
3. Siddhartha Mukherjee. The Song of The Cell. An Exploration of Medicine and the New Human. Scribner

Course and grading policy

Tentative grading and course policy:

Attendance and class participation = 10%

Quizzes = 20% (no retake in case you miss it)

Assignments = 30%

Exams = 40% (2 exams)

Mark distribution out of 100

Attendance = 10 marks

Quiz = 20 marks (2 quizzes- 10 marks each)

Assignment = 30 marks (2 assignments – 15 marks each)

Exams = 40 marks (20 + 20)

Attendance policy

Please inform the instructor if you plan to miss a class well in advance. In case of unavoidable circumstances like health or family emergency please send an e-mail

Please make it a point **show up to the class on time**

Attendance **less than 60% will be considered Fail**